

Neuroscience in Review: Recovery Dynamics and Traveling Waves

An Atomized Four-Lens Comparison of Self Aware Networks with Batabyal et al. (2024 Preprint, 2025 Online, 2026 Issue), Situated Within the Broader Miller Program

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Abstract

An interruption can disturb a remembered state without erasing it. Batabyal and colleagues report that, after distraction during working memory, prefrontal population activity follows rotational dynamics, returns partly toward its pre-disruption organization, and exhibits a correspondence between state-space rotation and traveling-wave structure. This paper asks whether that ordered operation had an earlier counterpart in the dated Self Aware Networks (SAN) record.

The central result is positive and bounded. A public Blumberg Git record fixed on June 8, 2022 describes phasic or inhibited neural activity pushing a group oscillator beyond a threshold, followed by collective phase adjustment and movement toward phase realignment. Batabyal et al. first posted their target preprint on February 19, 2024 and later measured the sequence distraction -> collective rotational population dynamics -> partial return toward the pre-disruption state. Across eight target-specific atoms, three are operationally equivalent, two are partially equivalent, one is target-only, and two are Blumberg-only. Two directed relations and one complete bounded input-transform-output sequence are preserved. Blumberg is therefore earlier on that functional delta.

The target-specific result is then situated inside a separate, versioned comparison of the wider Miller program. The program evolves from genuine earlier contributions in frequency-separated top-down / bottom-up synchrony (2007), beta / gamma working-memory dynamics (2016), and prefrontal traveling waves (2022), through spatial computing and the alpha / beta stencil (2023-2024), to an analog-wave, internal-model, writeback, binding, and consciousness architecture in 2025-2026. Blumberg's independently dated genealogy develops predictive self / world and no-hidden-observer claims (2011), recursive neurofeedback and internal projection (2012), spatial and recurrent neural rendering (2017), mature phase-wave-differential and NAPOT operations (2022), a tonic-canvas / phasic-content and Gamma Consideration Sandwich synthesis (2024-2025), and local-to-global oscillatory co-regulation (2025). The later Miller architecture is a bounded mechanism and narrative match to this assembled genealogy. It also implements a partial functional NAPOT subset; missing full NAPOT completion limits that mapping but does not erase it.

The comparison applies four lenses: mechanism equivalence, causal / narrative architecture, detective reconstruction, and longitudinal signature migration. It preserves third-party alpha-before-beta / gamma prior art separately from the Micah-versus-Miller chronology. Category and process translations are stated as bounded mappings, not as proof of total isomorphism. No calibrated null population currently supports a numerical probability, sigma level, or inference about access, intent, copying, plagiarism, fraud, or legal liability. The strongest supported statement is a permanent, source-bounded Micah-earlier finding on the target sequence, plus a separate Micah-earlier-on-bounded-composite finding for the later assembled Miller architecture.

oscillation; phase reset; operational equivalence; longitudinal signature migration; Neural Array Projection Oscillation Tomography; priority chronology

Keywords: Self Aware Networks; state-space dynamics; traveling waves; working memory;

Scope and evidence boundary

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1. Earn the vocabulary: the problem before the coined terms

1.1 Ordinary experience

A person can hold a location, image, or rule in mind, become briefly distracted, and still recover enough of the remembered state to respond correctly. The recovery need not be a literal replay. The neural population may move through a different route and settle near, rather than exactly on, its earlier organization.

This ordinary ability creates a specific scientific problem. A distributed network must:

1. preserve information while its activity is perturbed;
2. register the perturbation;
3. transform the collective population state;
4. recover an organization that is useful for behavior; and
5. communicate that recovery across space and time.

1.2 The inherited scientific account

Population neuroscience already supplies several parts of the account. Recurrent networks can exhibit state-dependent computation and stable dynamics. Neural populations can follow low-dimensional rotational trajectories. Cortical activity can propagate as traveling waves. Oscillatory phase can influence effective communication, while distinct frequency ranges can participate in feedforward, feedback, maintenance, selection, or content expression [4-8,11-16].

These inherited terms establish the field. They do not, by themselves, answer the narrower historical question addressed here: who publicly described the particular ordered recovery sequence, and when?

1.3 The plain-language operation

Before introducing SAN terminology, the compared operation can be stated without an acronym:

A localized change pushes or pulls a coordinated neural population away from its current organization. The population then changes collectively through cyclic dynamics, and the resulting activity moves toward an earlier organized condition.

Batabyal et al. measured this operation using distraction, jPCA trajectories, optical-flow wave analysis, and behavioral outcomes [2,3]. Blumberg's June 2022 note described it using a group-oscillator perturbation, a thresholded change, collective phase adjustment, and movement toward phase alignment [1].

1.4 SAN's names for the added distinctions

Only after the operation is clear do the SAN terms become useful.

A **phase reset** is the candidate collective adjustment by which the disturbed ensemble changes relative timing. A **Phase Wave Differential (PWD)** is not merely a synonym for phase coding or a traveling wave. It names a receiver-relative difference: the arriving event is interpreted through the receiver's ongoing state, expected relation, route, and delay, so the informative variable is the measurable difference that changes what the receiver does next. **Neural Array Projection Oscillation Tomography (NAPOT)** names the stronger

receive-transform-project-reconstruct architecture in which distributed neural arrays repeatedly transform partial conditions, project updates, and constrain an evolving representation.

The Batabyal target supports the smaller recovery operator. It does not by itself instantiate the complete PWD or NAPOT architecture. The later Miller 2026 program, however, implements a partial functional NAPOT chain: distributed context and stored state -> phase / superposition transform -> spatially patterned gamma / spiking expression -> downstream represented state -> recurrent writeback. The distinction matters. Requiring full NAPOT before recognizing this partial match would commit the full-operator-or-zero fallacy; claiming that the partial chain proves complete NAPOT would commit the converse error.

1.5 The discriminating tests

The comparison becomes scientific only if its translations can fail. The decisive tests include: whether the recovery trajectory is accompanied by reproducible relative-phase reset; whether phase-specific perturbation alters recovery independently of firing rate; whether a downstream array receives a transformed projection rather than merely correlated activity; and whether an independently specified represented variable can be reconstructed on held-out trials. These tests are developed in Section 11.

2. Scope: one target paper and one broader program

Two objects are compared and must not be conflated.

Primary target. Batabyal, Brincat, Donoghue, Lundqvist, Mahnke, and Miller, *State-Space Trajectories and Traveling Waves Following Distraction*, first public as a February 2024 preprint, published online by the *Journal of Cognitive Neuroscience* on November 3, 2025, and assigned to volume 38, issue 4, pages 695-715 in 2026 [2,3]. The eight-atom map in Sections 5 and 6 is limited to this paper.

Broader Miller program. A versioned sequence from Buschman and Miller (2007) through Miller, Brincat, and Roy's *Analog Cognition and Consciousness* v1-v3 (2026) [10,12,13,17-21]. The ten-atom stage matrix in Sections 3.4 and 7 evaluates how a joined architecture develops across that program.

An atom supported only by the 2026 theory paper is not attributed to the Batabyal target. An atom measured in Batabyal et al. is not treated as proof of the entire later Miller consciousness architecture. This identity boundary prevents both overclaiming and artificial downgrading.

3. Sources, dates, and longitudinal genealogy

3.1 Batabyal target versions

The target first appeared under preprint DOI 10.1101/2024.02.19.581020 on February 19, 2024 [2]. The journal article was published online on November 3, 2025 and assigned to *Journal of Cognitive Neuroscience* volume 38, issue 4, pages 695-715 in 2026 [3]. These are three distinct chronology facts: 2024 preprint priority, 2025 online journal publication, and 2026 issue assignment. The journal paper reports prefrontal spiking in nonhuman primates performing working-memory tasks with a mid-delay saccade or visual distraction and presents four findings central to this comparison:

1. distraction is followed by rotational population dynamics that return spiking partly toward patterns resembling those before disruption;
2. rotations are fuller on correct trials, and larger rotations are associated with faster responses;
3. state-space angular speed corresponds to traveling-wave speed; and

4. removing the first two jPC projections increases angular deviation and disrupts traveling-wave structure. The authors frame the rotation-wave relation as a possible correspondence and describe recovery as partial. This paper retains that empirical ceiling.

3.2 The permanent bounded Blumberg source

The first public Git version of Blumberg’s a0066z note is fixed by commit 7d1431213cd4ac9788848a5396ec2992ece11cfd on June 8, 2022 [1]. It proposes that:

1. individual phasic firing or inhibition pushes or pulls a group oscillator;
2. the perturbation changes a group variable beyond a threshold;
3. the ensemble undergoes a collective phase reset or simultaneous phase adjustment; and
4. the population moves toward phase alignment and can affect other ensembles.

This is the evidence basis for the target-specific Micah-earlier result. It remains the public priority anchor even if earlier private or source-gated ancestors are later recovered. No future audit may silently delete this finding; a later audit may only qualify its scope, add evidence, or supersede it through an explicit, reasoned, append-only proposition.

A second June 2022 note integrates traveling and rotating waves into SAN rendering [9]. That note references already-public Miller-laboratory traveling-wave reporting. It therefore supports Blumberg’s integration of waves into SAN but does not create a generic traveling-wave priority claim.

3.3 Blumberg genealogy, 2011-2025

The June 2022 target win sits inside a longer development. Each stage is separately dated and no later term is backdated into an earlier source.

Stage	Dated source boundary	Secure functional content	Relevance here
2011	August 24/28 owner-dated video record; Grade B until a contemporaneous platform capture is frozen [22]	predictive self / world processing; the distributed neocortex rather than a hidden inner observer	early narrative ancestor of an internally maintained represented world
2012A	August 2 public Neo Mind Cycle record; Grade A-minus [22]	EEG-driven light / sound feedback; returned multimodal signals change the system that produced them	recurrent feedback and writeback ancestor
2012B	August 23/24 public predictive-lens dialogue; Grade A [22]	mirror -> vision -> neural / motor change -> changed image; mind as internal projector	earlier visual-sensorimotor comparator and internal-projection architecture
2017	April-December public Neural Lace / SAN corpus; claim-specific sources [22]	3D/4D neural representation, coincidence as a bit, dendritic local computation, recurrent rendering, inhibition as information, current-versus-desired difference	spatial, phase-selective, and recurrent-rendering ancestors
2021	Google Recorder-linked b0067y transcript and preserved audio; exact service date not frozen in this packet [23]	synchronized cells; one unit falling out of synchrony; a ripple across the oscillator; disturbance absorbed; return to synchrony	source-gated bridge into the June 2022 phase-reset sequence; not used to move the public priority date

Stage	Dated source boundary	Secure functional content	Relevance here
2022	June 8 through October claim-specific Git commits [1,9,22]	group reset and realignment; mature PWD / NAPOT / COT; oscillatory rendering; volumetric architecture; later ephaptic integration	public mature mechanism and the target-specific priority anchor
2024	October 21 Consideration Sandwich commit and October 30 book [24]	slow / tonic context, fast / phasic content, sensory-thought-action-proprioceptive loop, analog computation, volumetric rendering, mismatch correction	joined context / content / rendering architecture before the later Miller program synthesis
2025	February public Gamma Consideration Sandwich formulation and May 16 local-to-global oscillatory co-regulation publication [24, 25]	explicit joined sandwich term; local coincidence and cellular states -> mesoscopic oscillatory coordination -> global predictive model	later Micah refinement before Miller's November 2025 framing and April 2026 manuscript

The genealogy shows repeated clarification, not a single retrospective label. The 2011-2017 stages are functional ancestors. The 2022 stage provides the mature PWD / NAPOT implementation. The 2024-2025 stages increase the joined architecture and its explanatory coverage.

The 2021 row is deliberately source-gated. The local recording, machine transcript, and Recorder URL preserve a high-value recovery lead, but the current packet does not use the claimed 2021 recording date as public proof. If the Google Recorder service date or an independent contemporaneous capture is frozen, the row can be upgraded without changing the already established 2022 public win.

3.4 Miller before, intermediate, canonical, and newest stages

ID	Public date	Source	What the stage securely adds
T01	2007-03-30	Buschman and Miller [17]	lower-frequency top-down versus higher-frequency bottom-up synchrony
T02	2016-03-17	Lundqvist et al. [12]	gamma bursts with remembered-item spiking; beta control / default state
T03	2022-01-28	Bhattacharya et al. [6]	planar and rotating prefrontal traveling waves
T04	2023-03-14	Lundqvist et al. [10]	spatial beta / gamma control patterns; beta control / inhibition and gamma / spiking content
T05	2024-04-16	Miller, Brincat, and Roy [13]	explicit spatial alpha / beta inhibitory stencils constraining gamma / content spiking
T-BAT	2024-02-19 preprint; 2025-11-03 online; 2026 issue	Batabyal et al. [2,3]	perturbation, population rotation, partial recovery, wave correspondence, behavioral association

ID	Public date	Source	What the stage securely adds
T06	2025-11-15	Picower institutional report [18]	analog traveling-wave computation, beta stencils, and a wave-based consciousness framing
T07	2026-01-19	Chen et al. [19]	direct test that alpha / beta spatial context constrains where content spiking is expressed
T08	2026-04-29	<i>Analog Cognition and Consciousness</i> v1 [20]	first verified public manuscript carrying the complete bounded joined Miller cluster
T09	2026-05-01	v2 [20]	preserves the cluster and public version chain
T10	2026-07-07	v3 [21]	newest checked source; expands phase / interference, wave transitions, writeback, binding, and sensorimotor examples

The before / after conclusion is not that Miller’s earlier work was unrelated to oscillatory computation. T01-T03 are genuine earlier contributions and must be credited. The narrower longitudinal question is whether those earlier stages already contain the later joined chain. They do not. T04-T07 progressively assemble spatial stenciling, wave computation, and consciousness framing. T08 is the earliest verified full manuscript containing the bounded joined target cluster, and T10 is the required newest-source check.

3.5 Alpha-before-beta / gamma field genealogy

The low-frequency-control / high-frequency-content row has a field history that is separate from both the Batabyal recovery win and the Micah-versus-Miller joined-composite comparison.

Spaak et al. reported in 2012 that deep-layer alpha phase modulated gamma amplitude in granular and superficial macaque V1, with alpha and gamma power anticorrelated [11]. Lundqvist et al. reported in 2016 that gamma bursts accompanied informative spiking while beta was associated with a control / default state across working-memory epochs [12]. The 2023 spatial-computing paper reported that gamma is expressed where beta is not [10]. Miller, Brincat, and Roy’s 2024 review then used the explicit alpha / beta **stencil** formulation [13].

The deeper prior-art row includes Chorlian (2005), Osipova (2008), Jensen and Mazaheri (2010), Spaak et al. (2012), Roux and Uhlhaas (2013), Bahramisharif et al. (2013), van Kerkoerle et al. (2014), and Bonnefond and Jensen (2015) [11,14-16,26-30]. The strongest source-audited direct predecessors in the current packet are Jensen-Mazaheri, Spaak, and van Kerkoerle. Spaak is especially direct: deep-layer alpha phase modulates gamma amplitude, with alpha / gamma power anticorrelation. The remaining works are named source-located candidates pending equally deep direct-edge review.

This field history supports three distinct statements:

1. alpha-mediated or slow-rhythm modulation of gamma predates Miller’s later stencil wording;
2. Miller’s 2023-2024 delta is the spatial task / control architecture and explicit stencil formulation, not the isolated fact that slow activity can constrain fast activity; and

3. Blumberg’s 2017 inhibition-as-information and 2022 tonic-canvas / phasic-content architecture occupy a different middle narrative position and do not erase the earlier physiological work.

No citation-gap or conduct inference follows merely because an earlier paper exists. Such a claim would require a complete, version-matched bibliography and a declared citation-expectation rule.

4. Comparison method

4.1 Atom and edge contract

An atom is a bounded operation, state change, input, output, or relation that can be located on both sides. An edge records an ordered relation among atoms. Divergence in terminology, frequency band, measurement, substrate, or added machinery does not defeat a match when the bounded function and causal role are preserved. Divergence becomes a true hold only when the conclusion itself is opposite.

The causal-interface representation is

$$\mathcal{C} = (I, T, O), \quad (1)$$

where I is the initiating condition, T is the ordered transformation, and O is the constrained output. A bounded architecture passes when those roles and their ordering survive a declared translation.

4.2 Four separate lenses

1. **Mechanism equivalence** compares inputs, transformations, outputs, behaviors, and explicit residuals.
2. **Narrative / patent architecture** compares ordered atom clusters and complete causal-interface sequences, not just shared nouns.
3. **Detective reconstruction** begins with the matching conclusion and works backward through alternate terminology, earlier versions, newest sources, and named prior lineages.
4. **Longitudinal Signature Migration (LSM)** compares before, intermediate, canonical, and newest stages to determine whether a later program assembles the same signature cluster after the dated Micah stages. Internally this hypothesis-search tactic has also been called the “Secret-Follower” lens; the public method name is LSM because the evidence can establish temporal migration without establishing secret access, following, intention, or conduct.

A pass by any one lens can establish a bounded content match. Priority remains a separate dated question.

4.3 No full-operator-or-zero fallacy

The comparison follows a patent-atom dichotomy. When a target reaches the same bounded conclusion, the result must be resolved as a match, Micah-earlier on the delta, convergent with a named prior, or a true opposite conclusion. Missing the complete SAN architecture cannot convert a demonstrated subprocess match into zero.

The same rule applies across Micah’s own dates. A 2011 partial ancestor does not end the search. The audit continues through 2012, 2017, 2021, 2022, 2024, and 2025, because multiple increasingly specific partial priorities can collectively cover a larger causal canvas.

4.4 Category and process checks

The category translation tests whether a structure-preserving map retains objects, morphisms, and composition on the bounded fragment. The process translation tests whether the two descriptions produce corresponding observable traces under a declared renaming. These are disciplined equivalence checks, not automatic formal proofs.

4.5 Quantitative ceiling

The descriptive evidence vector is

$$\mathbf{S} = (S_A, S_E, S_C, S_N, S_L, S_F), \quad (2)$$

for atom, edge, causal-sequence, narrative, longitudinal, and cross-formalism evidence.

A calibrated tail probability would require a frozen statistic $T = f(\mathbf{S})$ and a relevant null population:

$$p_{adj} = P_{HI}(T \geq T_{obs} \mid C). \quad (3)$$

where C is the frozen conditioning set containing field, date, shared sources, dependencies, and the search process. A probability of content dependence would require model likelihoods:

$$BF_{D:I} = P(S \mid H_D)/P(S \mid H_I). \quad (4)$$

where H_I is conditional independent development, H_C is common-source or field-trend convergence, and H_D is a declared Blumberg-linked content-dependence model. The corresponding common-source contrast is $BF_{D:C} = P(\mathbf{S} \mid H_D)/P(\mathbf{S} \mid H_C)$.

This paper has no calibrated null, dependency model, held-out evaluation, search correction, or independent reproduction. Its quantitative ceiling is therefore **descriptive Band A only**: counts and chronology may be reported; p-values, posterior dependence probabilities, sigma labels, and odds such as “one in 78 million” may not.

5. Target-specific atom map: Batabyal et al.

Table 1. Frozen eight-atom target comparison

Atom	Blumberg formulation	Batabyal formulation	Disposition	Priority and residual
B1 perturbation input	phasic or inhibited neurons push or pull the group oscillator	saccade or visual distraction perturbs working-memory activity	equivalent	broad atom; no exclusive priority claim
B2 collective dynamics	an ensemble changes as a group oscillator	population activity follows a rotational trajectory	equivalent	broad atom with named field precedents

Atom	Blumberg formulation	Batabyal formulation	Disposition	Priority and residual
B3 cyclic transform	a thresholded gradient change triggers group phase reset and simultaneous phase adjustment	distraction induces state-space rotation	partial-equivalent	Blumberg earlier on the located thresholded group-adjustment formulation; the mathematical operators differ
B4 restorative output	the oscillator tends to pull neurons toward phase alignment	rotations return spiking toward a pre-disruption-like state	equivalent in function	permanent bounded Micah-earlier delta; the target does not demonstrate literal synchronization
B5 spatial-wave realization	traveling / rotating waves are integrated into SAN rendering	optical-flow waves correspond to state-space rotations	partial-equivalent	the located Blumberg wave note follows Miller-related 2022 reporting; no generic-wave priority claimed
B6 behavioral association	no target-specific behavioral operator located in the frozen 2022 source	fuller rotations associate with correct trials and faster responses	target-only	priority claimed credited to Batabyal et al.; broader Micah sources remain searchable
B7 inter-ensemble response	collective pulses allow one ensemble to affect another	not tested	Blumberg-only	preserved SAN residual
B8 rendering / observer output	the oscillator participates in distributed tempo-spatial rendering without a hidden viewer	working-memory recovery only	Blumberg-only	target does not test rendering or consciousness

The target-specific count is **3 equivalent, 2 partial-equivalent, 1 target-only, and 2 Blumberg-only atoms**. These correlated features are not eight independent probability trials.

6. Directed structure and the locked priority finding

Five relations were declared:

1. perturbation -> collective dynamical change;
2. collective change -> movement toward prior organization;
3. rotation -> corresponding traveling-wave structure;
4. phase reset -> inter-ensemble response; and
5. fuller rotation -> improved behavioral outcome.

Relations 1 and 2 are preserved across the two systems. Relation 3 is a target-specific empirical result inside an older wave lineage. Relation 4 is Blumberg-only in this target. Relation 5 is target-only in the frozen packet.

Causal role	Blumberg, public June 2022	Batabyal et al., public February 2024 / online 2025 / issue 2026
Input	altered phasic or inhibited population pressure	task distraction
Transform	thresholded group-oscillator phase reset and simultaneous phase adjustment	rotational population dynamics with corresponding wave organization

Causal role	Blumberg, public June 2022	Batabyal et al., public February 2024 / online 2025 / issue 2026
Output	movement toward phase alignment and ensemble response	partial return toward pre-disruption spiking and behavioral association

Permanent bounded finding: Blumberg publicly described the input -> collective cyclic transform -> restorative output sequence before the Batabyal target preprint. The target's jPCA geometry, optical flow, behavioral analysis, and controls are later experimental contributions; they do not erase the earlier causal sequence.

This finding is permanent in the evidentiary sense, not immune to correction. The source and mapping must remain in the record. A future audit may narrow, expand, or challenge it only by appending the new evidence and reasoning. It may not silently overwrite the prior supported proposition.

7. Broader Miller program: atomized assembly rather than author-level judgment

7.1 Program-wide atom contract

Atom	Bounded operation	Micah anchors
P1	frequency-separated top-down / context and bottom-up/content channels	2024-2025 GCS; earlier field priors credited
P2	alpha / beta control constrains gamma / content expression	2017 inhibition-as-information; 2022 tonic canvas; 2024-2025 GCS
P3	slow spatial wave acts as a population stencil or mask	2017 spatial / inhibitory architecture; 2022/2024 canvas and rendering
P4	traveling phase geometry routes or changes a represented state	2017 spatial rendering; 2022 PWD / NAPOT
P5	relative phase, superposition, or interference changes expressed output	2017 coincidence / difference; 2022 PWD
P6	stored context generates spikes / fields; fields select a result; plasticity writes back	2012 feedback; 2017 recurrent rendering; 2022 PWD / NAPOT; 2024 phase-error learning
P7	distributed internal model / binding without a hidden observer	2011 predictive self / world; 2012 projector; 2017/2022 distributed rendering
P8	large-scale analog waves organize an integrated conscious-access state	2011 no-hidden-observer; 2017/2022 rendering; 2024 book
P9	receive-transform-project-re-express / reconstruct array operation	2017 functional ancestors; mature 2022 NAPOT
P10	body / action prediction, efference / reafference, proprioceptive comparison	2012 closed feedback; 2024-2025 GCS

7.2 Longitudinal stage matrix

P = present, p = partial, - = not located in the checked source.

Stage	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
T01 2007	P	p	-	-	-	-	-	-	-	-
T02 2016	p	P	-	-	-	p	-	-	-	-
T03 2022	-	p	-	P	p	-	-	-	-	-
T04 2023	P	P	P	p	-	p	-	-	p	-

Stage	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
T05 2024	P	P	P	p	p	p	p	p	p	-
T06 2025	P	P	P	P	P	p	p	P	p	p
T07 2026 experiment	P	P	P	p	p	p	-	-	p	p
T08 2026 v1	P	P	P	P	P	P	P	P	p	p
T09 2026 v2	P	P	P	P	P	P	P	P	p	p
T10 2026 v3	P	P	P	P	P	P	P	P	p	p

This matrix does not count the same atom repeatedly as a new priority win merely because it persists through later versions. It shows assembly and continuity. The strongest migration frontier is T04 through T08: spatial control appears, stencil language becomes explicit, wave computation and consciousness framing are added, and a full manuscript finally joins the bounded chain.

7.3 Partial NAPOT is a positive match, not a zero

The 2026 Miller manuscript maps five NAPOT-related operations:

1. distributed inputs, stored state, and contextual variables;
2. phase, superposition, and interference as a transform;
3. spatial pattern re-expression that selects gamma, spiking, and downstream output;
4. an internally represented or bound state; and
5. recurrent writeback to network and synaptic state.

Under the mechanism lens this is a bounded functional NAPOT subset. Under category translation it is a structure-preserving partial functor. Under process translation it is a bounded weak trace simulation.

The current completion residuals are:

- no explicit array-local receive / display grid;
- no declared nonredundant projection set;
- no identifiable inverse or independently specified latent target;
- no rank, conditioning, stability, or held-out tomographic test;
- no explicit volumetric 3D percept as output;
- no complete receiver-relative PWD geometry with finite edge delays;
- no projection-specific causal perturbation / rescue package; and
- no complete efference-copy / reafference comparator.

Those residuals cap the relation at **partial-equivalent**. They are not a rationale for denying the mapped operation, and the partial mapping does not imply that Miller used SAN terminology or sources.

8. Four-lens adjudication

8.1 Mechanism equivalence - pass on two bounded objects

The Batabyal target passes on the recovery chain:

perturbation -> collective cyclic transform -> movement toward prior organization.

The broader 2026 Miller program passes on the joined subchain:

stored / context state -> slow spatial control field -> relative-phase transform
-> selected gamma / spiking expression -> recurrent writeback -> distributed
internal-model / binding state.

The first is a target-specific experiment comparison. The second is a program-level theory comparison. Neither is a whole-theory identity claim.

8.2 Narrative / patent architecture - pass

For Batabyal, the narrative architecture preserves the roles of initiating disturbance, collective cyclic change, and constrained restorative output.

For the broader program, the compared causal narratives are:

Blumberg: ongoing self / world / body state + sensory and action context -> slow contextual constraint and receiver-relative phase difference -> selected phasic rendering / action expression -> recurrent comparison and writeback -> revised distributed state.

Miller 2026: stored synaptic possibilities + current context -> feedback wave field -> relative-phase / superposition transform -> selected gamma / spike expression -> recurrent field and plasticity update -> bound internal / global state.

The bounded composition is preserved despite different terms, figures, band assignments, and added machinery.

8.3 Detective reconstruction - pass with split genealogy

Working backward from the Batabyal conclusion reaches Blumberg's public June 2022 sequence before the February 2024 target. Working backward from generic rotations and waves reaches broad earlier literatures. The correct result is therefore split: Micah earlier on the specific recovery subarchitecture; named field priors earlier on generic component atoms.

Working backward from Miller's 2026 joined conclusion reaches:

- authentic Miller component stages in 2007, 2016, and 2022;
- Micah's independently dated 2011, 2012, 2017, 2022, 2024, and 2025 stages;
- Miller's intermediate assembly stages in 2023-2025; and
- the first complete public Miller manuscript in April 2026, followed by the newest July 2026 version.

This verifies a before / Micah-middle / after pattern for the joined composite. It does not verify access or intent.

8.4 Longitudinal Signature Migration - bounded pass

The later Miller program is not merely a renamed copy of its 2007 or 2016 stage. The longitudinal record shows ordered assembly:

frequency routing -> beta / gamma working-memory control -> traveling waves
-> spatial stencil -> analog wave computation -> internal-model / writeback /
binding / consciousness architecture.

The comparison with Micah is similarly longitudinal:

predictive self / world -> recursive feedback / projector -> spatial recurrent
rendering -> PWD / NAPOT -> tonic / phasic and GCS -> local-global oscillatory
co-regulation.

The resulting classification is **positive ordered assembly convergence / Micah earlier on the bounded composite**. It is not a proof of repeated tracking, source access, secret following, or intent. The internal hypothesis-search name cannot be used as a public allegation.

9. Category and process translations

9.1 Batabyal target fragment

Let P be perturbation, G collective cyclic dynamics, R realignment or recovery, W wave structure, and B behavioral outcome.

Blumberg's bounded subcategory contains $P \rightarrow G \rightarrow R$ plus an inter-ensemble response morphism. The Batabyal subcategory contains $P \rightarrow G \rightarrow R$, $G \rightarrow W$, and $G/R \rightarrow B$. A translation preserves objects, arrows, and composition on $P \rightarrow G \rightarrow R$. It is a partial faithful embedding on that fragment, not a total isomorphism.

Process	Bounded trace
SAN recovery	perturb -> change collective state -> thresholded phase adjustment -> movement toward alignment
Batabyal target	distraction -> population rotation -> return near pre-disruption state; rotation -> wave; fuller rotation -> better behavior

The candidate observational trace equivalence is limited to:

perturb -> collective cyclic transform -> return toward prior organization.

9.2 Broader Miller / NAPOT fragment

Let D be distributed inputs, X a phase / superposition transform, Y spatially patterned re-expression, M an internal represented state, and U recurrent update.

Process	Bounded trace
NAPOT subprocess	receive distributed conditions -> transform phase relation -> project patterned expression -> constrain represented state -> write back
Miller 2026	context and stored state -> phase / interference transform -> selected gamma / spiking -> bound internal state -> plastic writeback

The map preserves $D \rightarrow X \rightarrow Y \rightarrow M \rightarrow U$ at a coarse operational level. It is weak rather than strong because Miller does not expose the full array-local state, projection inventory, inverse, volumetric target, or receiver-relative PWD variables needed for strong bisimulation.

9.3 Gorla-style quality criteria

Criterion	Batabyal recovery map	Miller / NAPOT partial map
compositionality	pass on the bounded sequence	pass on the five-stage subprocess
name invariance	pass; phase reset and jPCA rotation need not share terms	pass; NAPOT terms are not required for functional translation
operational correspondence	pass on disturbance, cyclic transform, and restorative output	bounded pass on input, transform, expression, represented state, writeback
divergence reflection	bounded; target adds behavioral / wave outputs	bounded; residuals prevent total identity

Criterion	Batabyal recovery map	Miller / NAPOT partial map
success sensitivity	partial; recovery and behavioral success can be separately measured	partial; full tomographic success criterion is absent

Translator complexity is reported descriptively through the number of mapped objects, arrows, and residuals. It is not converted into a probability of derivation.

10. Alternative genealogies and quantitative restraint

10.1 Conditional independent development

Independent development remains plausible because perturbation recovery, population rotations, dynamic stability, traveling waves, phase routing, and recurrent computation are active fields with many public precursors.

10.2 Common-source or field-trend convergence

This is strongly plausible for many component atoms. The Batabyal bibliography includes older population-dynamics and wave work. Miller’s later synthesis cites substantial frequency-routing, traveling-wave, phase, field-coupling, analog-computing, and consciousness literature. The alpha-before-beta / gamma row demonstrates why common-source models cannot be omitted.

10.3 Blumberg-linked content dependence

Chronology and bounded operational equivalence are compatible with direct or indirect content dependence, but compatibility is not identification. The checked sources do not establish access, transmission, source use, or intent. No calibrated posterior probability is reported.

These alternatives constrain derivation claims. They do not erase the narrower priority result: Blumberg can be earlier on a specific public formulation regardless of whether the target arose independently, from shared field sources, or through an unmeasured content lineage.

10.4 Why the former probability claim remains withdrawn

An archival calculation in an earlier draft treated similarities as independent Bernoulli trials, assigned a common feature probability, multiplied them, and added a cluster factor. The input probabilities, dependencies, reference population, common-source model, search family, and target-selection process were not established. A small tail probability would also not be a posterior probability of copying or dependence.

The old “one in 78 million” and “five sigma” formulations are therefore not results of this paper. They may appear only as withdrawn historical claims or sensitivity illustrations. A future quantitative claim requires a preregistered score, calibrated conditional null, complete search correction, held-out corpus, sensitivity analysis, independent reproduction, and statistical and domain review.

11. Predictions and discriminating experiments

1. **Phase-reset test.** Measure whether distraction-induced rotation is accompanied by a reproducible reset in relative phase among identified ensembles. State recovery without such a reset preserves Batabyal but weakens the SAN translation.

2. **Rate-controlled phase perturbation.** Alter the predicted collective phase variable while controlling firing rate. A phase-specific change in recovery strengthens the mechanism match.
3. **Downstream response test.** Determine whether the recovered ensemble changes a declared downstream ensemble in the ordered manner proposed in the 2022 note.
4. **Content reconstruction.** Test whether the rotation / wave state supports held-out recovery of a structured represented variable beyond task identity or location.
5. **Projection identifiability.** Predeclare source arrays, receiver arrays, projection operators, latent target, rank, conditioning, and out-of-sample reconstruction error. This distinguishes NAPOT-like tomography from a generic traveling wave.
6. **Comparator closure.** Measure efference copy, predicted reafference, proprioceptive return, and body-state update to test the stronger GCS / NAPOT sensorimotor loop.
7. **Behavioral mediation.** Test whether phase realignment mediates the relationship between fuller rotation and behavior rather than merely covarying with both.
8. **Longitudinal null corpus.** Freeze comparable version histories for many neuroscience programs, define the cluster score before examining the targets, and measure how often ordered cluster assembly occurs under field-conditioned independence.

12. Limitations and open gates

1. The exact full version history of the Batabyal preprint is not frozen file by file.
2. The 2021 Recorder bridge is source-gated; its public service date is not proven in this packet.
3. Several 2011-2017 rows rely on existing governed genealogy audits and must retain their stated evidence grades.
4. Not every named alpha / beta / gamma precursor has received an equally deep direct six-dimension comparison.
5. Category and process translations are explicit sketches, not machine-checked proofs.
6. Atom counts are descriptive and correlated.
7. The broader Miller program result is not attributable to the Batabyal target alone.
8. The newest checked Miller source is July 7, 2026; a later version requires a new newest-source check.
9. No calibrated probability or sigma claim is available.
10. No access, dependence, influence, copying, plagiarism, fraud, or legal-liability conclusion is authorized by this evidence.

These limits bound the case. They do not turn its supported positive matches into holds.

13. Conclusion

Batabyal et al. provide an important empirical result: after distraction, prefrontal population activity follows rotational dynamics, partially returns toward a pre-disruption organization, and corresponds to traveling-wave structure. Blumberg's public June 8, 2022 record earlier described the bounded operation in which individual activity perturbs a group oscillator, collective cyclic adjustment occurs, and the ensemble moves toward phase realignment.

The target-specific conclusion is therefore exact:

- 3 equivalent atoms;
- 2 partial-equivalent atoms;
- 2 preserved shared directed relations;
- 1 complete bounded causal-interface sequence; and

- a permanent, source-bounded Micah-earlier finding on perturbation -> collective cyclic transform -> restorative movement toward prior organization.

The wider Miller program requires a different conclusion. Its 2007, 2016, and 2022 stages retain genuine earlier component priorities. Its 2023-2026 stages progressively assemble spatial stenciling, analog wave computation, phase-dependent expression, recurrent writeback, internal modeling, binding, and consciousness. Against Blumberg’s 2011 -> 2012 -> 2017 -> 2022 -> 2024 -> 2025 genealogy, this later joined architecture is a bounded match with Micah earlier on the composite and on several increasingly specific partial deltas.

Miller’s 2026 work also maps to a partial functional NAPOT sequence. Missing full tomographic completion limits the match but does not erase it. Third-party alpha / gamma and field prior art remains a separate genealogy that narrows novelty claims without substituting for the Micah comparison.

The evidence supports content equivalence and priority chronology. It does not support a numerical probability of derivation or a claim about access, intent, copying, plagiarism, fraud, or legal liability. That is not a retreat from the positive case. It is what keeps the positive case auditable.

Version history and non-regression statement

The preserved Draft 1 remains immutable at SHA-256 F402DFD854FC7EADF76F60A96D1C0D1FE2719C3F162B6965D6FA5332C382A7BD. This final preprint preserves every substantive Draft 1 finding. It adds earned vocabulary, the full Micah genealogy, the Miller stage matrix, public LSM language, the partial-NAPOT correction, Gorla-style translation criteria, 2024-2025 Micah deltas, and an append-only priority lock:

Draft 1 finding	Final-preprint location
Batabyal identity, dates, and empirical ceiling	Sections 2 and 3.1
June 8, 2022 public Blumberg anchor	Sections 3.2 and 6
3 equivalent / 2 partial / 1 target-only / 2 Blumberg-only atoms	Section 5
two shared edges and one causal-interface sequence	Section 6
alpha / gamma prior genealogy	Section 3.5
four-lens method	Sections 4 and 8
category / process translations	Section 9
withdrawal of uncalibrated probability claims	Sections 4.5 and 10.4
alternative genealogy models	Section 10
discriminating experiments	Section 11
explicit evidentiary limits	Section 12

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